

Question 1

Objective

Decide which algorithm to use to label parcel boundaries as FRONT, SIDE, or REAR. This is a needed step to apply appropriate setbacks to define buildable areas.

Attempts

There are 2 algorithms outlined in the Evidence section. One can only handle 4-side parcels, and the other can handle parcels of any shape but it is probably more difficult to implement.

Evidence

The 2 algorithms being compared are described in the report linked below. The dependencies, pros, cons, and steps for each are included, along with a diagram. Focus on the following sections of the report:

- Labelling Boundary Lines for 4-Sided Parcels
- Labelling Boundary Lines for Parcels of Any Shape

https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/blob/main/code/gis_files/reports/buildable_area_algorithms.pdf

Diagnosis

There are many approaches to the buildable area problem, each with their advantages.

Proposed Next Steps

Implement the more robust algorithm that can support parcels of any shape.

Decision Ask

Please review the 2 algorithms and let us know if you see any pitfalls with the robust algorithm that would warrant implementing the algorithm that is limited to only 4-sided parcels instead.